Data Flow Diagram

Data Flow Diagram Introduction:

A Data Flow Diagram (DFD) visually maps out how information moves through a system—detailing the interactions between external entities, automated processes, and database storage modules. The DFD diagram on the next page traces the exact lifecycle of credit applications from browser-based parameter submission through feature engineering, model classification matching accuracy scoring, and database override filtration rules.

DFD Symbol Legend:

Symbol	Name	Description / Project Implementation mapping
Oval / Rounded shape	**External Entity**	A person, role, or external system that sends or receives data parameters. (e.g., Credit Analyst inputting applications, Prospective Customer check wizard, Compliance Auditor CSV files).
Rectangle with numbered header	**Process**	An operational function that transforms incoming data values into output variables. (e.g., Process 1: Request Preprocessing, Process 2: Model Inference, Process 3: Compliance Checking).
Rectangle (solid border, no number)	**Data Store**	A persistent registry cache or directory file containing data for lookup. (e.g., preprocessor.pkl encoding weights, model.pkl classifier model, session Evaluation History Log).
Labeled arrow	**Data Flow**	Arrows indicating the movement direction of data vectors between entities, processes, and stores, labeled explicitly with what parameters are being transferred.

System Architecture Flow Example:

On the following page, our credit validation pipeline is detailed in a Level 1 Data Flow Diagram illustrating how variables move from the frontend browser widgets through the Flask API routing logic, matching model storage inference algorithms, and saving to log databases.

Level 1 Data Flow Diagram: Credit Card Approval Prediction System